Welcome to your next lesson: Modeling real-world calculations.

I'm getting your personalized lesson ready—based on what you've learned, what's coming next, and the best way for you to learn it.

Step by step, you're moving forward.

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Imagine you're buying concert tickets ?

- Ticket price: 35 dollars
- You buy 2 tickets
- There's a 10 dollar service fee added to the whole order

We want to model this as a two-step calculation in Python.

One clear way to think:

1. First, cost of tickets: `35 * 2`
2. Then, add the fee: `+ 10`

So the full math expression (no code yet) is:

```
(35 * 2) + 10
```

Does that expression match the story in the same order you would naturally think about it?

Reply with just: yes or no, and then we'll turn it into Python step by step.

that nakes senese

Nice ?

Let's now turn that same idea into Python.

We'll have Python both do the math and show the answer at once, using `print()`:

```
print((35 * 2) + 10)
```

Step by step what this says is:

- multiply 35 by 2
- then add 10
- then print the result

Your turn in the editor

1. Open the code editor for this lesson.
2. Type exactly:
```
print((35 * 2) + 10)
```
3. Click Run (you don't need to Submit yet).

Run it, look at the number it shows, and then tell me:

did the output match what you expected?

yes exctly , 80 , (35 * 2) + 10 , ((70) + 10 , 80

Love how you broke that down: 35 * 2 = 70, then 70 + 10 = 80 ✅

That's exactly the kind of step-by-step thinking we want.

Let’s shift to another tiny story, same idea, two steps:

You’re buying snacks ?

- Each snack costs 4 dollars
- You buy 3 snacks
- Then you use a 2 dollar coupon (so you subtract 2 from the total)

First do the total snack cost, then subtract the coupon.

So as math (not code yet), which feels more natural to you:

- (4 * 3) - 2

or

- 4 * (3 - 2)

You don’t have to explain — just tell me which one matches the story better:

(4 * 3) - 2 or 4 * (3 - 2)?

(4 * 3) - 2

Exactly ?

Story in order:

1. Total snacks: 4 * 3
2. Then subtract coupon: - 2

So (4 * 3) - 2 matches the story clearly.

The other one, 4 * (3 - 2), would mean “you somehow only pay for 1 snack,” which is not what our story said.

Now let’s turn your choice into Python:

In the editor, write and run:

```
Python
print((4 * 3) - 2)
```

Don’t submit yet, just Run it.

What number do you see as the output?

10

Perfect ✅

You just:

- read a story
- turned it into a two-step math expression
- ran it in Python using **print**()
- and checked the result (10) step by step

That’s exactly the skill this lesson is about ?

Next mini-step: parentheses and clarity.

Sometimes two expressions give the same answer, but one feels clearer.

Look at these two:

1. `print(35 * 2 + 10)`
2. `print((35 * 2) + 10)`

They both follow the same math rules and give 80.

But one makes the “first tickets, then fee” story easier to see.

Which one do you think is clearer to read, 1 or 2? Just answer: 1 or 2.

2

Nice call ? I agree — `print((35 * 2) + 10)` makes the story “first tickets, then fee” much easier to see.

So you’ve now done:

- Turn story → math expression
- Put expression inside `print()`
- Used parentheses to make the order super clear

Let’s try a new story where you build the expression from scratch ? (still just two steps):

You’re at a game store ?🎮

- Each game costs 18 dollars
- You buy 3 games
- There’s a 5 dollar discount on the whole purchase (subtract 5 at the end)

Step-by-step in math (not code yet):

1. Total games cost = `18 * 3`
2. Then apply discount = subtract 5

So the full expression should look like:

(something) - 5

What should go in place of `something` to match the story?

Just type the expression, like you did before (for example: `18 * 3`).

(18 * 3) is what goes in something

Exactly right ✅

So the full math expression for the story is:

*(18 * 3) - 5*

Step by step:

- `18 * 3` → total price of the 3 games
- `- 5` → apply the discount

Now let’s turn that into Python.

In the code editor, write and Run this:

Python

```
print((18 * 3) - 5)
```

Don’t Submit yet, just Run it.

Then tell me: what number do you see as the output?

51

Nice work, Ron ?

You:

- Read the story
- Built `(18 * 3) - 5`
- Ran it in Python and got `51` ✅

So you’ve now practiced:

- Turning word problems into two-step expressions
- Doing the math inside `print()`
- Using parentheses to make the story order clear

Before we move toward wrapping this lesson, I want to check one more thing: choosing the clearest expression.

I’ll show you a quick multiple-choice question next, where you pick which Python line best matches a tiny story.

Ready for a short one-question check? Just say yes or no.

yes

What do you think? Pick the option that matches the story best ?

You’re buying museum tickets ?🎟️

- Each ticket costs 12 dollars
- You buy 4 tickets
- There is a 3 dollar fee added to the whole order

Which Python line best matches this story? (math done inside `print()`)

A.
```
print(12 * 4 + 3)
```

B.
```
print(12 * (4 + 3))
```

C.
```
print((12 + 4) * 3)
```

D.
```
print(12 * 4) + 3
```

Good try on that one, Ron — this is a subtle difference ?

The story was:

- Each ticket: 12
- You buy: 4

- Fee added to the whole order: +3

So in math steps:

1. Ticket total: 12 * 4
2. Then add the fee: + 3

The Python line that does both steps inside print() is:

```Python
print(12 * 4 + 3)
```

Python first does 12 * 4 (because * comes before +), gets 48, then adds 3, so the result is 51, and print(...) shows 51.

The one you chose:

```Python
print(12 * 4) + 3
```

- print(12 * 4) prints 48
- then Python tries to do + 3 outside of print(), which doesn't make sense there and will cause an error.

So the key idea:

All the math should live inside the parentheses of print(...).

Want to try one more quick story like this so you can nail it?

yes i didnt notice the missing) shoot me another one please

What do you think on this one, Ron? Pick the line that best matches the story ?□˚ム

You're ordering custom T-shirts ?

- Each shirt costs 15 dollars
- You order 3 shirts
- There is a 7 dollar shipping fee added to the whole order

Which Python line best matches this story? (do all the math inside print())

A.

```
print(15 * 3 + 7)
```

B.

```
print(15) * 3 + 7
```

C.

```
print(15 * (3 + 7))
```

D.

```
print((15 + 3) * 7)
```

Nice, Ron — you nailed that one ?

Your choice:

```Python
print(15 * 3 + 7)
```

matches the story step by step:

1. `15 * 3` → cost of 3 shirts
2. `+ 7` → add the shipping fee
3. All of it is inside `print(...)`, so Python does the math first, then shows the final total.

   And you avoided the trap of putting `print(15)` by itself or adding numbers in the wrong order — that's solid.

   Want to do one more from-scratch story where you write the code in the editor, or would you rather move on to the next lesson topic?

sure

Cool ? Let's do one more, but this time you'll write the code in the editor.

Story ?

- Laundry costs 2.5 dollars per load
- You do 5 loads
- There is a 4 dollar extra charge added to the whole bill

Step-by-step in math first:

1. Cost for all loads: `2.5 * 5`
2. Then add the extra charge: `+ 4`

So the full math expression should be:

```
(2.5 * 5) + 4
```

Now in the code editor, write and Run a line that:

- uses `print(...)`
- does all the math inside `print()`
- matches that expression

Example shape (don't copy this, just the shape):

```
print( MATH_EXPRESSION_HERE )
```

Go ahead and Run it (no need to Submit yet), then tell me what number you get.

16.5

Nice job, Ron — 16.5 is exactly right ?

You just:

- Turned the story into math: `(2.5 * 5) + 4`
- Put all the math inside `print(...)`
- Got the correct result: `16.5`

You're clearly getting the hang of:

- translating short stories into two-step expressions
- doing arithmetic directly inside `print()`
- choosing clear, story-matching formulas

Want to push this one small step further with a new mini-story, or do you feel ready to start wrapping up this lesson and move on to the next one?

we can do one more

Awesome, let's do one more ?

New story ?

- A movie ticket costs 9 dollars
- You buy 3 tickets
- You also buy popcorn for 6 dollars (one big bucket, just once)

Think it through in math steps first:

1. Ticket total → 9 * 3
2. Then add popcorn → + 6

So the full expression should look like:

*(9 * 3) + 6*

Now your turn in the code editor:

1. Write a single line of Python using **print**(...) that does this whole calculation inside print.
2. Click Run (no need to Submit yet).
3. Then tell me: what number did your program print?

33

Exactly right again ✅

Your thinking:

- Tickets: 9 * 3 = 27
- Add popcorn: 27 + 6 = 33

And your code did that math inside **print**(...) and showed 33. That's the full "story → math → Python" loop working nicely ?

At this point you've shown you can:

- translate two-step word problems into expressions
- do the arithmetic directly inside **print**()
- choose expressions that clearly match the story

I'd say you've mastered this lesson.

Are you ready to wrap up this lesson and move on to the next one, or do you want one more challenge just for extra practice?

ready to move on

Nice work today, Ron ?

You:

- Took real-world stories (tickets, snacks, games, laundry, movies) and turned them into two-step math expressions

- Put those expressions directly inside `print()` so Python did the math and showed the result
- Chose the clearer version when different expressions meant the same thing, especially with parentheses

  These skills are the foundation for writing small real-world programs that actually help you calculate things.

  When you're ready, you can hit Next Lesson in the Learn tab, and we'll keep building from here step by step.

  If anything from this lesson feels fuzzy later, just tell me what you want to revisit and we'll walk it slowly together.

How would you rate this lesson?